1. Bit Stuffing

AIM Write a program to implement Bit Stuffing.

CODE

```
#include <stdio.h>

int main()
{
    int a[100], b[200];
    int n, i, m=0, c=0;

    printf("Enter length of the given string: ");
    scanf("%d", &n);

    printf("Enter the bits:\n");
    for(i=0; i<n; i++)
    {
        scanf("%d", &a[i]);
    }

    for(i=0; i<n; i++)
    {
        b[m] = a[i];
        m++;

        if(a[i] == 1)
        {
            c++;
        }
        else
        {
            c = 0;
        }

        if(c == 5)
        {
            b[m] = 0;
            m++;
            c = 0;
        }
    }

    printf("\nAfter bit stuffing:\n");
```

```
    for(i=0; i<m; i++)
    {
        printf("%d ", b[i]);
    }
    printf("\n");

    return 0;
}
```

OUTPUT

Plaintext

```
Enter length of the given string: 8
Enter the bits:
1 1 1 1 1 0 1 0

After bit stuffing:
1 1 1 1 1 0 0 1 0
```

2. Character Stuffing

AIM Write a program to implement Character Stuffing.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char s1[100], s2[100];
    char s3[100] = "DLESTX";
    char s4[40] = "DLEETX";

    printf("Enter string: ");
    scanf("%s", s1);

    strcpy(s2, s3);

    int j = strlen(s2);
    int i = 0;

    for(i=0; i<strlen(s1); i++)
    {
```

```
        if(s1[i] == 'D' && s1[i+1] == 'L' && s1[i+2] == 'E')
        {
            s2[j] = 'D';
            s2[j+1] = 'L';
            s2[j+2] = 'E';
            j = j+3;
        }

       s2[j] = s1[i];
        j++;
    }

    s2[j] = '\0';
    strcat(s2, s4);

    printf("\nStuffed String: ");
    puts(s2);

    return 0;
}
```

OUTPUT

```
Enter string: HELLODLEWORLD

Stuffed String: DLESTXHELLODLEDLEWORLDDLEETX
```

3. Fixed Framing

AIM Write a program to implement Fixed Framing.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char st[100];
    int n, i, j, c=0, count=0, size;

    printf("Enter the string: ");
    scanf("%s", st);

    printf("Enter the no of frames: ");
    scanf("%d", &n);

    int frames[n];

    printf("Enter the frame size: ");
    scanf("%d", &size);

    for(i=0; i<n; i++)
        frames[i] = size;

    printf("\nThe no of frames: %d\n", n);

    for(i=0; i<n; i++)
    {
        printf("\nThe content of the frame %d: ", i);

        j=0;

        while(c < strlen(st) && j < frames[i])
        {
            printf("%c", st[c]);

            if(st[c] != '\0')
                count++;

            c = c+1;
            j++;
        }
```

```
        printf("\nsize of frame %d : %d\n", i, count);
        count = 0;
    }

    return 0;
}
```

OUTPUT

```
Enter the string: COMPUTERNETWORKS
Enter the no of frames: 4
Enter the frame size: 4

The no of frames: 4

The content of the frame 0: COMP
size of frame 0 : 4

The content of the frame 1: UTER
size of frame 1 : 4

The content of the frame 2: NETW
size of frame 2 : 4

The content of the frame 3: ORKS
size of frame 3 : 4
```

4. Variable Framing

AIM Write a program to implement Variable Framing.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char st[100];
    int n, i, j, c=0, count=0;

    printf("Enter the string: ");
    scanf("%s", st);
```

```
    printf("Enter the no of frames: ");
    scanf("%d", &n);

    int frames[n];

    printf("Enter the frame size of the frames:\n");
    for(i=0; i<n; i++)
    {
        printf("Frame %d: ", i);
        scanf("%d", &frames[i]);
    }

    printf("\nThe no of frames: %d\n", n);

    for(i=0; i<n; i++)
    {
        printf("\nThe content of the frame %d: ", i);

        j=0;

        while(c < strlen(st) && j < frames[i])
        {
            printf("%c", st[c]);

            if(st[c] != '\0')
                count++;

            c = c+1;
            j++;
        }

        printf("\nsize of frame %d : %d\n", i, count);
        count = 0;
    }
    printf("\n");
    return 0;
}
```

OUTPUT

```
Enter the string: NETWORKING
Enter the no of frames: 3
Enter the frame size of the frames:
```

```
Frame 0: 3
Frame 1: 4
Frame 2: 3

The no of frames: 3

The content of the frame 0: NET
size of frame 0 : 3

The content of the frame 1: WORK
size of frame 1 : 4

The content of the frame 2: ING
size of frame 2 : 3
```

5. Checksum

AIM Write a program to implement Checksum.

CODE

```
#include <stdio.h>

int sender(int arr[], int n)
{
    int checksum, sum = 0, i;
    printf("\n****SENDER SIDE****\n");
    for(i = 0; i < n; i++)
        sum += arr[i];

    printf("SUM IS: %d", sum);
    checksum = ~sum; // 1's complement of sum
    printf("\nCHECKSUM IS: %d", checksum);

    return checksum;
}

void receiver(int arr[], int n, int sch)
{
    int checksum, sum = 0, i;
    printf("\n\n****RECEIVER SIDE****\n");
    for(i = 0; i < n; i++)
        sum += arr[i];
```

```
    printf("SUM IS: %d", sum);
    sum = sum + sch;
    checksum = ~sum; // 1's complement of sum
    printf("\nCHECKSUM IS: %d\n", checksum);
}

int main()
{
    int i, n, sch, rch;

    printf("\nENTER SIZE OF THE STRING: ");
    scanf("%d", &n);

    int arr[n];

    printf("ENTER THE ELEMENTS OF THE ARRAY TO CALCULATE
CHECKSUM:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    sch = sender(arr, n);
    receiver(arr, n, sch);

    return 0;
}
```

OUTPUT

```
ENTER SIZE OF THE STRING: 3
ENTER THE ELEMENTS OF THE ARRAY TO CALCULATE CHECKSUM:
10
20
70

****SENDER SIDE****
SUM IS: 100
CHECKSUM IS: -101

****RECEIVER SIDE****
SUM IS: 100
CHECKSUM IS: 0
```

6. CRC

AIM Write a program to implement CRC.

CODE

```
#include <stdio.h>
#include <string.h>

char data[100], div[100], temp[100], total[100];
int i, j, datalen, divlen, len, flag=1;

void check();

int main()
{
    printf("Enter the total bit of data: ");
    scanf("%d", &datalen);

    printf("\n Enter the total bit of divisor: ");
    scanf("%d", &divlen);

    len = datalen + divlen - 1;

    printf("\n Enter the data: ");
    scanf("%s", data);

    printf("\n Enter the divisor: ");
    scanf("%s", div);

    for(i=0; i<datalen; i++)
    {
        total[i] = data[i];
        temp[i] = data[i];
    }

    for(i=datalen; i<len; i++)
        total[i] = '0';

    check();

    for(i=0; i<divlen-1; i++)
        temp[datalen+i] = data[i];
```

```
    temp[datalen+divlen-1] = '\0'; // Added to prevent
printing garbage values

    printf("\n Transmitted Data: %s\n", temp);

    // Fixed the smart quotes to standard quotes here
    printf("\n\n Enter the received code word: ");
    scanf("%s", total);

    check();

    for(i=0; i<divlen-1; i++) {
        if(data[i] == '1') {
            flag = 0;
            break;
        }
    }

    if(flag==1)
        printf("\n successful\n");
    else
        printf("\n received code word contains errors...\n");

    return 0;
}

void check()
{
    for(j=0; j<datalen; j++)
    {
        if(total[j]=='1')
        {
            for(i=0; i<divlen; i++)
                total[j+i] = (total[j+i] == div[i]) ? '0' :
'1';
        }
    }

    for(i=0; i<divlen-1; i++)
        data[i] = total[datalen+i];
    data[i] = '\0';
}
```

OUTPUT

```
Enter the total bit of data: 4

 Enter the total bit of divisor: 4

 Enter the data: 1001

 Enter the divisor: 1011

 Transmitted Data: 1001110


 Enter the received code word: 1001110

 successful
```

7. Go Back N

AIM Write a program to implement Go Back N.

CODE

```
#include <stdio.h>

int main()
{
    int w, i, f, frames[50];

    printf("Enter window size: ");
    scanf("%d", &w);

    printf("Enter no of frames to transmit: ");
    scanf("%d", &f);

    printf("Enter %d frames: ", f);
    for(i=1; i<=f; i++)
        scanf("%d", &frames[i]);

    printf("\nWith sliding window protocol the frames will be
sent in the following manner (assuming no corruption of
frames)\n");
```

```
    printf("After sending %d frames at each step sender waits
for acknowledgement sent by the receiver\n", w);

    for(i=1; i<=f; i++)
    {
        if(i%w==0)
        {
            printf("%d\n", frames[i]);
            printf("Acknowledgement of above frames sent is
received by the sender\n");
        }
        else
        {
            printf("%d ", frames[i]);
        }
    }

    if(f%w!=0)
    {
        printf("\nAcknowledgement of above frames sent is
received by the sender\n");
    }
    return 0;
}
```

OUTPUT

```
Enter window size: 3
Enter no of frames to transmit: 5
Enter 5 frames: 10 20 30 40 50

With sliding window protocol the frames will be sent in the
following manner (assuming no corruption of frames)
After sending 3 frames at each step sender waits for
acknowledgement sent by the receiver
10 20 30
Acknowledgement of above frames sent is received by the
sender
40 50
Acknowledgement of above frames sent is received by the
sender
```